MILITARY SPECIFICATION

GROMMETS, RUBBER, SPLIT, GENERAL PURPOSE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of split rubber grommets.

1.2 These grommets are intended for use as a protection, from abrasion or vibration, for rods, wires, tubes, and cables passing through walls, bulkheads and partitions. The grommets can also be used for applications where contact with oil, grease or gasoline may occur at temperature below 150° F.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and publications of the issue in effect on the date of invitation for bid form a part of this specification.

SPECIFICATIONS

FEDERAL

TT-E-527 — Enamel Synthetic Lustreless

TT-E-529 — Enamel Synthetic Semi-gloss

TT-E-489 — Enamel Synthetic Gloss

PPP-B-566 — Boxes, Folding, Paperboard

PPP-B-585 — Boxes; Wood, Wirebound

PPP-B-591 — Boxes; Fiberboard, Wood-Cleated

PPP-B-601 — Boxes, Wood, Cleated-Plywood

PPP-B-621 —Boxes; Wood, Nailed and Lock-Corner

PPP-B-636 — Boxes; Fiber

PPP-B-676 — Boxes, Set-Up, Paperboard

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MIL-B-4229 — Boxes, Paperboard, Metal-Stayed

MIL-R-003065 — Rubber and Synthetic Rubber Compounds Mechanical and General Purpose

MIL-B-10377 — Box, Wood, Cleated, Veneer, Paper Overlaid

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STANDARDS

FEDERAL

Federal Test Method Standard No. 601 — Rubber

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MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes

MIL–STD–129 — Marking for Shipment and Storage

MS35490 — Grommet, Rubber, Split, General Purpose

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Consolidated Freight Classification Rules and Container Specifications.

(Application for copies of the above publication should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

National Motor Freight Classification Rules and Container Specifications (Either No. 11 or No. A–1).

(Application for copies of the above publication should be addressed to Issuing Officer, American Trucking Association, 1424 16th St., N.W., Washington, D. C.)

3. REQUIREMENTS

3.1 Design. The design of the grommet shall conform to Standard MS35490. (See 2.1.)

3.2 Material. The grommets shall be made from synthetic rubber the physical properties of which shall conform to Specification MIL–R–003065A, grade No. SB–512ABF2.

3.3 Color. The color shall be black.

3.4 Flexibility. Grommets shall withstand insertion, at any temperature between 17.7° C. (0.0° F.) and 65.5° C. (150° F.) in a hole $\frac{1}{32}$ inch larger than dimension "C" shown on Standard MS35490.

3.5 Dimensions, sizes and tolerances. The dimensions and sizes of the grommets shall be in agreement with Standard MS35490. The permissable dimensional variations (tolerances) shall also be as shown on the standards.

3.6 Workmanship. The grommets shall be hand- or machine-trimmed, or they shall be tumbled with dry ice to remove flash. They should also be free of dirt, oil, or other foreign matter. There should be no bloom, mold marks, or other injurious defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot. A lot shall consist of all the grommets on a contract or order that are one size, manufactured under essentially the same conditions, and made of the same compound (see also 3.2) for delivery at one time to one location.

4.2 Sampling, inspection, and test procedures. These operations shall be in accordance with Standard MIL–STD–105 (including the appendix) and Specification MIL–R–003065A and shall be performed either by the government or the contractor under the supervision of a government inspector, as designated by the procuring agency.

4.2.1 *Sampling*

4.2.1.1 *For nondestructive inspection.* Sampling for nondestructive inspection shall be in accordance with Standard MIL–STD–105, Inspection Level II, single sampling plan.

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4.2.1.2 *For basic requirements tests.* Sampling for flexibility, tension and elongation tests shall be in accordance with appendix to Standard MIL–STD–105, Inspection Level L6 (also see par. 3.4).

4.2.2 *Inspection*

4.2.2.1 *Packaging, packing and marking.* Each lot shall be visually examined by the inspector to determine that the packaging, packing, and marking conform to this specification (also see sec. 5).

4.2.2.2 *Classification of defects.* Inspection of the grommets shall be conducted as follows:

Categories and defects	*Gauge or other method of inspection*
Major — *AQL 2.5 percent*	
1. Outside diameter	Machinists' rule
2. Depth of groove	Machinists' rule
3. Hole diameter	Inside caliper
Minor — *AQL 10.0 percent*	
51. Bloom, flash, mold mark and other injurious defects	Visual
52. Contamination, oil, dirt, or foreign matter	Visual

4.2.3 *Chemical composition.* The inspector shall make certain that the chemical composition of the grommet complies with paragraph 3.2. Unless otherwise specified, the rubber manufacturers' notarized certification of the chemical composition shall be sufficient evidence (see also par. 4.2.4).

4.2.4 *Test procedures.* Unless otherwise specified all tests shall be made on the compound from which the grommet will be made to assure that the composition of the compound shall conform to this specification and Specification MIL–R–003065 (see also 4.2.3). Disposition of the lot with respect to these tests shall be in accordance with the criteria for AQL 6.5 percent of Standard MIL–STD–105 for each test.

4.2.5 *Rejection and resubmittal.* If the total number of defective grommets is equal to or greater than the rejection number associated with the AQL values, the lot shall be rejected. A rejected lot may be resubmitted for inspection after the manufacturer has completed 100 percent inspection of the lot based on the deficiencies noted, and has removed all nonconforming units.

5. PREPARATION FOR DELIVERY

5.1 Packaging

5.1.1 *Level A*

5.1.1.1 *Unit packaging.* Grommets of like size and description shall be packaged in quantities of 100 for ¾-inch or less outside diameter and quantities of 50 or less for outside diameters greater than ¾-inch. The grommets shall be packaged in close fitting paperboard folding boxes, paperboard set-up boxes or metal edge boxes conforming to Specifications PPP–B–566, PPP–B–676 or MIL–B–4229.

5.1.1.2 *Intermediate packaging.* The grommets of like size and description, unit packaged as specified in 5.1.1.1, shall be intermediate packaged in close fitting fiber boxes conforming to Specification PPP–B–636, class 2 or 3. Gross weight of intermediate containers shall not exceed 20 pounds.

5.1.2 *Level B.* The grommets, unit packaged as specified in 5.1.1.1, shall be intermediate packaged as specified in 5.1.1.2, except that the containers shall conform to class 1 of Specification PPP–B–636.

5.1.3 *Level C.* The grommets shall be packaged to afford adequate protection against deterioration and physical damage in accordance with the manufacturer's standard practice.

5.2 Packing

5.2.1 *Level A.* The grommets, packaged as

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specified in 5.1.1 shall be packed in closed-fitting nailed wood, wood cleated plywood, wirebound wood, wood cleated fiber or wood cleated veneer paper overlaid boxes conforming to Specifications PPP–B–621 (class 2) PPP–B–601 (overseas type), PPP–B–585 (class 3), PPP–B–591 (overseas type) and MIL–B–10377 (overseas type). When specified, boxes conforming to specification PPP–B–636 (class 3) may be used. The boxes shall be closed and strapped in conformance with the appendix to the applicable box specification. The gross weight of wood boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification.

5.2.2 *Level B.* The grommets, packaged as specified in 5.1.2, shall be packed in close-fitting nailed wood, wood cleated plywood, wirebound wood, or wood cleated fiber boxes conforming to Specifications PPP–B–621 (class 1), PPP–B–601 (domestic type), PPP–B–585 (class 1), and PPP–B–591 (domestic type). When specified, boxes conforming to class 1 of Specification PPP–B–636 may also be used. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.3 *Level C.* The grommets, packaged as specified in 5.1.3, shall be packed to insure carrier acceptance and safe delivery to destination. Containers shall comply with Consolidated Freight Classification Rules and Container Specifications or with National Motor Freight Classification Rules and Container Specifications (No. 11 or No. A–1) or with other carrier regulations applicable to the mode of transportation. The gross weight of each container shall not exceed 200 pounds.

5.3 Marking

5.3.1 *Normal marking.* Marking of packages and shipping containers shall be in accordance with Standard MIL–STD–129.

5.3.2 *Special marking*

5.3.2.1 *Level B.* Interior packages prepared for Level B and shipping containers containing these packages shall be conspicuously marked "LTD–MIL–PKG".

5.3.2.2 *Level C.* Shipping containers containing grommets packaged for Level C shall be conspicuously marked "MIN–MIL–PKG".

5.3.2.3 *Protection of overseas exterior markings.* Overseas exterior container markings shall be protected by a coating of transparent waterproof material as specified in paragraph 3.2 MIL–STD–129. This protection is not required for service color corner markings, for lithographed containers and for containers marked with paint conforming to Specifications TT–E–527, TT–E–529 and TT–E–489.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Size as shown on MS35490

(1) Hole diameter

(2) Groove diameter

(3) Outside diameter

(4) Groove width

(c) Selection of applicable levels of packaging and packing. (See also par. 5.1 and 5.2)

(d) Weight limitations if greater than specified in section 5.

(e) Special markings, if required.

6.2 This specification adequately describes the characteristics and tests necessary to

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secure the desired material, without samples prior to award, to assure compliance with this specification. However, should, for any purpose, samples with bid be necessary, they should be specifically requested in the invitation for bid.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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